

# Syllabus

2026 Fall

|                          |                                   |                      |                     |
|--------------------------|-----------------------------------|----------------------|---------------------|
| <b>Course</b>            | Deep learning for computer vision | <b>Professor</b>     | Jeong Hong          |
| <b>Course No</b>         | SOC4180                           | <b>Class No</b>      |                     |
| <b>Schedule</b>          | MON4,5,6,THU9,10,11(B-101)        | <b>Grading Eval.</b> | Relative Evaluation |
| <b>Other Information</b> |                                   |                      |                     |

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| <b>Profile</b>           | <ul style="list-style-type: none"> <li>- PhD, EECS (Electrical Engineering and Computer Science), MIT (Massachusetts Institute of Technology), USA</li> <li>- MS, EE, KAIST (Korea Advanced Institute of Science and Technology), Korea</li> <li>- BS, EE, SNU (Seoul National University), Korea</li> <li>- Professor, EE, BJTU (Beijing Jiaotong University), China</li> <li>- Professor, EE, POSTECH (Pohang University of Science and Technology), Korea</li> <li>- Professor, EE, KNU (kyungbook National University), Korea</li> <li>- Visiting Scientist, EE, USC (University of Southern California), USA</li> <li>- Consultant, Dialog Systems, Lucent Bell Labs at Murray Hill, USA</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Objectives</b> | <p>This course introduces students to robotics and robot learning from first principles, progressing from the basic anatomy of a robot to autonomous locomotion policies. Students build, control, and analyze a full-scale humanoid ? the Unitree G1 ? entirely within the MuJoCo physics simulator, learning to read a robot model, compute its kinematics, design controllers, and ultimately train policies that replace those controllers.</p> <p>The first half of the semester is classical. Students study rigid-body transforms and coordinate frames, forward and inverse kinematics, contact and ground reaction, the support polygon and Zero Moment Point criterion, and the Linear Inverted Pendulum Model. By the fourth week they make a 29-degree-of-freedom humanoid walk using only geometry, a linear differential equation, and inverse kinematics, with no learning of any kind. Actuation, PD control, oscillator-based gaits, and state estimation from onboard sensors complete the classical foundation.</p> <p>The second half turns to learning. Students formulate locomotion as a Markov decision process, implement policy gradients and PPO, study reward shaping and its failure modes, and train locomotion policies using GPU-parallel simulation. Domain randomization, robustness evaluation, and the sim-to-real gap are treated directly, and vision-based perception and imitation learning extend the work toward current practice. Because students have already built a working controller by hand, they can measure precisely what learning adds and what it costs.</p> <p>The course is conducted entirely in simulation and requires no physical hardware. Each week pairs a structured lecture with a laboratory notebook that runs in Google Colab with zero installation, so every student works with the same robot on identical software. Through this integrated progression, students gain a solid foundation in classical robot control, modern robot learning, and the engineering judgment to know which one a problem calls for.</p> |
|                          | <p><b>Core Components</b></p> <p>The five-layer robot stack: physics, model, middleware, control, and autonomy<br/> Robot description formats (MJCF, URDF) and the body tree as a kinematic chain<br/> Degrees of freedom, floating-base systems, actuators, and sensors<br/> Control timescales, from 500 Hz physics to sub-hertz task decisions<br/> Kinematics and Geometry</p> <p>Rigid-body transforms, rotation matrices, quaternions, and axis-angle<br/> Forward kinematics of articulated chains, validated against a physics engine<br/> Jacobians and differential kinematics<br/> Inverse kinematics: closed-form solutions and damped least squares<br/> Singularities, joint limits, and workspace analysis<br/> Dynamics, Contact, and Balance</p> <p>Contact modeling, friction, and ground reaction forces<br/> Center of mass, center of pressure, and the support polygon<br/> The Zero Moment Point criterion for postural stability<br/> Underactuation, and why a legged robot cannot be commanded directly<br/> Classical Control and Analytic Locomotion</p> <p>PD and position control, gain tuning, and torque saturation<br/> The Linear Inverted Pendulum Model and its closed-form solution<br/> Footstep planning and boundary value problems for gait generation<br/> Swing-foot trajectories and double-support phases<br/> Central pattern generators as model-free gait synthesis</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|----------------------------|------------------------------------------------------------------------------------------|------------|-------------|------|------------|-----|-------|
| Course Description         | Sensing and State Estimation                                                             |            |             |      |            |     |       |
|                            | Inertial measurement: gyroscopes, accelerometers, and gravity as a tilt reference        |            |             |      |            |     |       |
|                            | Complementary filtering for body orientation                                             |            |             |      |            |     |       |
|                            | Observation design, and why policies observe gravity in the body frame                   |            |             |      |            |     |       |
|                            | Reinforcement Learning                                                                   |            |             |      |            |     |       |
|                            | Markov decision processes, policies, value functions, and returns                        |            |             |      |            |     |       |
|                            | Policy gradients, actor-critic methods, and PPO                                          |            |             |      |            |     |       |
|                            | Environment design under the Gymnasium API, and residual action parameterization         |            |             |      |            |     |       |
|                            | Reward shaping, reward hacking, and diagnosis of failed training runs                    |            |             |      |            |     |       |
|                            | GPU-parallel simulation for large-scale locomotion training                              |            |             |      |            |     |       |
| Textbooks                  | Robustness and Sim-to-Real                                                               |            |             |      |            |     |       |
|                            | Domain randomization over mass, friction, and actuation                                  |            |             |      |            |     |       |
|                            | Actuator lag, latency, backlash, and sensor noise                                        |            |             |      |            |     |       |
|                            | Quantifying the reality gap by evaluating policies on a degraded simulator               |            |             |      |            |     |       |
|                            | Policy export and onboard deployment concepts                                            |            |             |      |            |     |       |
|                            | Perception and Imitation                                                                 |            |             |      |            |     |       |
|                            | Rendering RGB and depth from onboard cameras                                             |            |             |      |            |     |       |
|                            | Terrain-aware locomotion and height-map perception                                       |            |             |      |            |     |       |
|                            | Motion tracking and imitation from reference data                                        |            |             |      |            |     |       |
|                            | Tools and Practice                                                                       |            |             |      |            |     |       |
| Other Texts and References | MuJoCo and MuJoCo Menagerie; the Unitree G1 humanoid                                     |            |             |      |            |     |       |
|                            | Gymnasium, PyTorch, Stable-Baselines3, and JAX/MJX                                       |            |             |      |            |     |       |
|                            | Reproducible, pinned environments; every lab runs in Google Colab with zero installation |            |             |      |            |     |       |
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| Class Structure            | Lecture                                                                                  |            |             |      |            |     |       |
|                            | Labs                                                                                     |            |             |      |            |     |       |
|                            | Projects                                                                                 |            |             |      |            |     |       |
|                            | Midterm Exam                                                                             |            |             |      |            |     |       |
|                            | Final Exam                                                                               |            |             |      |            |     |       |
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| Notes                      | Any of the violation of the following will be subject to Failure:                        |            |             |      |            |     |       |
|                            | - Academic rule: 1/4 (8 times) Absent, unless permitted by me (not AA) beforehand        |            |             |      |            |     |       |
|                            | - Any of this: No Lab, No Project. No Midterm Exam, No Final Exam                        |            |             |      |            |     |       |
|                            |                                                                                          |            |             |      |            |     |       |
|                            | Course contents may be changed without notification during the semester.                 |            |             |      |            |     |       |
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| ABEEK                      |                                                                                          |            |             |      |            |     |       |
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| Grading                    |                                                                                          |            |             |      |            |     |       |
| Mid-term                   | Final exam                                                                               | Attendance | Assignments | Quiz | Discussion | ETC | Total |
| 30 %                       | 30 %                                                                                     | 5 %        | 30 %        | 0 %  | 0 %        | 5 % | 100 % |

| Syllabus |                      |                                                                                                                |       |
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| Week     | Content              | Class                                                                                                          | Notes |
| 1        | <b>Theme</b>         | Artificial Neurons                                                                                             |       |
|          | <b>Class Details</b> | Introduction to neurons, activation functions (ReLU, GELU), and weight initialization strategies (Xavier, He). |       |
|          | <b>Tests</b>         |                                                                                                                |       |
| 2        | <b>Theme</b>         | Optimization & Loss                                                                                            |       |
|          | <b>Class Details</b> | Loss functions (MSE, cross-entropy), gradient descent variants, and backpropagation mechanics.                 |       |
|          | <b>Tests</b>         |                                                                                                                |       |

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| 3  | Theme         | Neural Components                                                                                  |  |
|    | Class Details | Core modules: Softmax, residual connections, dropout, normalization (BatchNorm, LayerNorm).        |  |
|    | Tests         |                                                                                                    |  |
| 4  | Theme         | Embedding & Representation                                                                         |  |
|    | Class Details | Tokenization, word embeddings (Word2Vec, GloVe), dense vector spaces, and semantic encoding.       |  |
|    | Tests         |                                                                                                    |  |
| 5  | Theme         | Positional Encoding & Attention                                                                    |  |
|    | Class Details | Positional encoding (sinusoidal, learned), scaled dot-product attention, and multi-head attention. |  |
|    | Tests         |                                                                                                    |  |
| 6  | Theme         | Reinforcement Learning                                                                             |  |
|    | Class Details | MDPs, policy gradients, Q-learning, and RL applications in agentic systems.                        |  |
|    | Tests         |                                                                                                    |  |
| 7  | Theme         | Midterm Exam                                                                                       |  |
|    | Class Details |                                                                                                    |  |
|    | Tests         |                                                                                                    |  |
| 8  | Theme         | Classical Architectures                                                                            |  |
|    | Class Details | MLPs, CNNs, pooling layers, and ResNet-style residual networks.                                    |  |
|    | Tests         |                                                                                                    |  |
| 9  | Theme         | Sequence Models                                                                                    |  |
|    | Class Details | RNNs, LSTMs, GRUs, and modeling temporal dependencies.                                             |  |
|    | Tests         |                                                                                                    |  |
| 10 | Theme         | Transformer Architecture                                                                           |  |
|    | Class Details | Self-attention, encoder-decoder structure, BERT, GPT, and scalability.                             |  |
|    | Tests         |                                                                                                    |  |
| 11 | Theme         | Generative Models I                                                                                |  |
|    | Class Details | Autoencoders, VAEs, GANs?latent space modeling and adversarial training.                           |  |
|    | Tests         |                                                                                                    |  |
| 12 | Theme         | Generative Models II                                                                               |  |
|    | Class Details | Diffusion models, DDPM, and image synthesis via denoising.                                         |  |
|    | Tests         |                                                                                                    |  |
| 13 | Theme         | Multimodal Models                                                                                  |  |
|    | Class Details | CLIP, BLIP, vision-language fusion, and cross-modal embeddings.                                    |  |
|    | Tests         |                                                                                                    |  |
| 14 | Theme         | Agentic Frameworks                                                                                 |  |
|    | Class Details | MCP, A2A, Toolformer, ReAct, and multimodal agent orchestration.                                   |  |
|    | Tests         |                                                                                                    |  |
| 15 | Theme         | Final Exam                                                                                         |  |
|    | Class Details |                                                                                                    |  |
|    | Tests         |                                                                                                    |  |
| 16 | Theme         | Makeup                                                                                             |  |
|    | Class Details |                                                                                                    |  |
|    | Tests         |                                                                                                    |  |